Best Selling Solar Kits – System Architecture & End-to-End Flow

Detailed Engineering Specification & Operational Guide for Admin & Solar Store

EXECUTIVE SUMMARY & BUSINESS GOAL

Boss ke requirements ke anusar, SolarKits platform par Best Selling Kits ka pura management 100% interconnected aur organic sales database driven banaya gaya hai.

• Pehle problem: Kits manually ek-ek karke fill karni padti thi aur hardcoded hoti thi.
• Solution: Ab Best Seller Configuration tab me live bookings ke aadhar par real-time dynamic ranking aati hai, aur Website Configuration me wahi data 1-click se auto-populate hokar Solar Store par live show hota hai.
• Regional Intelligence: Desh (Country), State aur District teeno levels par sales rank aur demand volume alag-alag calculate hoti hai.

1. 3 Key Interconnected Modules (High-Level Architecture)

Module / Tab Name	Location in Admin Portal	Key Responsibility	Connected Source Collection
1. Location Setup	Settings > Location Setting > Setup Location	Manages Active States, Districts & Pincodes list	geolocation_level_1, geolocation_level_2
2. Best Seller Matrix	Solar Kit Config > Best Seller Configuration	Calculates live organic sales rank & handles overrides	pc_combo_kits, best_seller_kits, orders
3. Website Showcase	Website Config > Solar Store Website	Syncs kits to store homepage & toggles 8 card fields	website_configurations (solar_store_homepage)
4. Storefront App	Solar Store (Customer / EPC View)	Dual-mode Pincode/District finder & localized badges	Direct API to shop.handler.js & geo.handler.js

2. End-to-End System Flow Diagram

STEP-BY-STEP DATA FLOW LIFECYCLE

1. GEO MASTER: Setup Location tab me Admin active states aur pincodes upload/manage karta hai.
2. SALES ENGINE: Orders DB (epc_orders, delivery_orders) se real booking counts aggregate hote hain.
3. BEST SELLER MATRIX: Best Seller Configuration tab in sabhi ko jodkar Country / State / District ke hisab se #1, #2, #3 ranking decide karta hai.
4. AUTO-SYNC TO SHOWCASE: Website Configuration me "Auto-Sync" button se Best Seller kits direct form me auto-fill ho jaati hain.
5. VISIBILITY CONTROLS: Website tab me admin decide karta hai ki kaunse field (Price, Watt, Discount, Rating, Stock) customer ko show karne hain.
6. STOREFRONT DISCOVERY: Customer Pincode ya State/District daalta hai toh use exact us area ki best seller kits localized badge ke sath milti hain.

Data Models & Organic Database Ranking Algorithm

Detailed explanation of mathematical weighting and schema relationships

3. Database Schema Relationships

Collection Name	Database	Key Fields Used	Role in Flow
geolocation_level_1	geolocation_db	_id, name, country, is_active	Defines States (e.g. Gujarat, Andhra Pradesh)
geolocation_level_2	geolocation_db	_id, name, level_1, pincodes[]	Defines Districts & their 6-digit Pincodes
pc_combo_kits	india_solarshop_db	_id, name, capacity, price, kit_image	Master catalog of pre-configured solar kits
best_seller_kits	india_solarshop_db	combo_kit_id, state_id, district_id, priority, badge_text, is_manual_override	Stores admin priority overrides & custom badges
epc_orders / delivery_orders	india_solarshop_db	items[].kit_id, destination_pincode, status	Live transaction records for calculating real sales volume

4. The Organic Sales-Driven Ranking Formula

SYSTEM DECIDES RANKING (NO GUESSWORK)

System khud database se order volume calculate karta hai. Jab tak Admin manually kisi kit ko pin nahi karta, tab tak ranking ka rule ye hai:

1. Sales Volume (Units Booked) -> Highest to Lowest (Primary)
2. Orders Count -> Highest to Lowest (Secondary tie-breaker)
3. Total Revenue (paise / INR) -> Highest to Lowest (Tertiary tie-breaker)

• Agar Admin "Make #1" click karta hai ya Priority: 1 set karta hai, toh `is_manual_override = true` set ho jata hai aur Admin ki pasand top par rehti hai.
• Agar Admin "Reset" click karta hai, toh kit dobara natural organic sales rank par aa jaati hai.

5. Regional Demand Weighting (Why Ranks Differ By Region)

Filter Level	Active Scope	Demand Weighting & Local Solar Pattern	Typical #1 Ranked Kit
National (All India)	state_id = null	Aggregates entire country bookings. 3kW dominates nationwide due to ¹78,000 PM Surya	3 kW Complete Home Combo Kit
South India (State)	Andhra Pradesh, Kerala, TN	Smaller residential rooftops & compact coastal homes lead to high demand for 1kW and 2kW	2 kW (Rank 1) / 1 kW (Rank 2)
West India (State)	Gujarat, Rajasthan	High solar irradiation, large bungalows & industrial sheds drive heavy 3kW and	3 kW (Rank 1) / 5 kW Commercial
North India (State)	Delhi, Punjab, Haryana, UP	Balanced residential demand across 2kW & 3kW net-metering rooftop setups.	3 kW (Rank 1) / 2 kW (Rank 2)
District Level	e.g. Rajkot, Anantapur, Jaipur	Scaled down to local district booking density (9%-19% of national volume) with localized ranking.	Varies by district booking logs

Website Configuration Sync & Storefront Discovery

Card field visibility controls and multi-tier cascading resolution on solar shop

6. Website Configuration: Auto-Sync & Field Toggles

HOW DATA MOVES FROM BEST SELLER TAB TO WEBSITE CONFIGURATION

Website Configuration > Solar Store Website tab me Admin ke paas 2 power tools hain:

1. "Auto-Sync from Best Seller Configuration" Button:
Is button par 1-click karne se Best Seller matrix ki top ranked kits automatic roop se Product Card #1, #2, #3... me load ho jaati hain. Unka Name, Capacity, Selling Price, MRP, aur Badges khud fill ho jaate hain.

2. Individual Product Card Auto-Fill Dropdown:
Har product card ke upar ek dropdown diya gaya hai: "-- Choose Kit to Auto-Fill Card --". Yahan se kisi bhi particular kit ko select karne par sirf us card ki details turant populate ho jaati hain.

Card Field Toggle	State Key	Control Description	Default State
Watt / Capacity	showWatt	Displays "3 kW Kit" or capacity pill	ACTIVE (SHOW)
Brand & Category	showBrand	Displays "SolarKits Prime • On-Grid Kit"	ACTIVE (SHOW)
Selling Price	showSellingPrice	Displays active selling price in INR (¹79,498)	ACTIVE (SHOW)
MRP & Savings	showMrp	Displays strikethrough MRP & savings banner	ACTIVE (SHOW)
Discount %	showDiscount	Displays green badge (e.g. 26% OFF)	ACTIVE (SHOW)
Rating & Reviews	showRating	Displays star rating score and EPC review count	ACTIVE (SHOW)
Stock & Dispatch	showStock	Displays "In Stock • Fast Delivery" status	ACTIVE (SHOW)
Add to Cart CTA	showCta	Displays "Order Now" / "Add to Cart" action buttons	ACTIVE (SHOW)

7. Storefront Cascading Resolution Logic (Waterfall)

STOREFRONT RESOLUTION WATERFALL ON SOLAR STORE

Jab koi customer ya EPC Solar Store ke Homepage par aata hai, toh backend kits ko is priority me check karta hai:

1. DISTRICT LEVEL TAG: Agar Admin ne us particular district (e.g. Rajkot) ke liye Best Seller tag lagaya hai, toh sabse pehle wahi dikhega ("Ø=Ý% #1 Best Seller in Rajkot").
2. STATE LEVEL TAG: Agar district me koi specific tag nahi hai, toh State level ka tag dikhega ("&¡ High Demand in Gujarat").
3. ALL INDIA NATIONAL TAG: Agar state me bhi override nahi hai, toh All India National Best Seller dikhega ("& #1 Top Pick in India").
4. DYNAMIC ORGANIC FALLBACK: Agar koi bhi manual tag nahi laga hai, toh catalog ki top kits live sales volume ranking ke sath automatically show ho jaati hain.

API Reference Table & Technical Solutions

Complete endpoint contracts and technical fixes implemented

8. Core API Endpoints Reference

Method & Endpoint	Module / Controller	Query Parameters	Output Response Payload
GET /admin-api/best-seller-configs	best_seller_kits.handler.js	state_id, district_id, kit_type, search, status	Unified list of ranked kits with units booked, orders count, revenue &
POST /admin-api/best-seller-configs	best_seller_kits.handler.js	req_for=add (Body: combo_kit_id, state_id,	Creates or updates regional best seller tag with is_manual_override.
PATCH /admin-api/best-seller-configs/:id/toggle-status	best_seller_kits.handler.js	Body: state_id, district_id, combo_kit_id	Toggles kit active/inactive status specifically for that regional scope.
GET /shop/best-sellers	shop.handler.js	district_id, state_id, pincode	Returns curated or dynamic sales-ranked best seller combo kits for
GET /geo/lookup-pincode/:pincode	geo.handler.js	:pincode (6-digit)	Resolves Pincode to District ID, State ID, State Name, and Locality.
GET /admin-api/website-configs/solar_store_homepage	website_config.handler.js	page_key=solar_store_homepage	Fetches current showcase product cards and field visibility toggles.

9. Key Technical Fixes Implemented

SUMMARY OF BUGS FIXED & STABILIZED

1. ALL INDIA SCOPE LEAKAGE (Andhra Pradesh Bug): Pehle All India scope me "Make #1" click karne par state_id missing hone par backend aur frontend states[0] (Andhra Pradesh) assign kar dete the. Ab state_id null rehti hai aur Target Region me "ALL INDIA (NATIONAL)" display hota hai.
2. DISTRICT DROPDOWN ON ALL INDIA: District dropdown ko All India scope par bhi enable kiya gaya hai jisme state label ke sath sabhi active districts list hote hain (e.g. "Jaipur (Rajasthan)").
3. DUAL-MODE STOREFRONT FINDER: FeaturedProducts.jsx me 2 modes diye gaye hain: Direct 6-digit Pincode Jump aur Cascading State -> District -> Pincode dropdowns.
4. PINCODE DATA STRUCTURE HARMONIZATION: Geo schema me pincode string aur object dono types ko match karne ke liye flexible resolver implement kiya gaya.